Agentic Delegation and the Language Frontier

Quispe and Xu (2026): model, evidence, and an independent Lean audit

Marcelo Chávez
Artificial Intelligence and Economic Modeling

github.com/amchavezu/ai-03-quispe

The citation described an earlier draft, not arXiv v2

Citation received

Quispe (2026)
Coding Beyond Your Training: Claude Code and the Technological Frontier of Software Developers

$\neq$

Verified v2

Alexander Quispe and Kevin Xu
Agentic Delegation and the Language Frontier of Software Developers: A Model and Evidence from Claude Code on GitHub

Revised July 7, 2026

What changed

The received title is acknowledged in the v2 source as an earlier version; authorship, title, framing, and evidentiary scope differ.

Source: Quispe and Xu, arXiv:2605.25438v2, title page and submission history.

The paper asks whether AI expands what a developer can produce

Question

Can agentic coding assistants make unfamiliar-language opportunities viable?

Contribution

Agentic AI adds a new production mode: the agent executes while the developer specifies, decomposes, and verifies.

Frontier object

Languages in which the developer **ships work**

$\neq$ languages the developer has learned to code unassisted

Central distinction: observed production, knowledge, general productivity, and agentic delegation are different objects.

Source: Quispe and Xu, Sections 1 and 4; Remark "What the model does and does not claim."

The developer follows a menu-and-threshold rule

$$
\begin{aligned}
V^S &= \omega + s\mu - \frac{\rho s^2}{2\pi} - b, \\
V^C &= V^S + \gamma s - r_C, \\
V^D &= \omega + (1-\lambda)s\mu + \lambda a z(A) - \kappa - r_D \\
&\quad - \frac{\rho}{2}\left[\frac{(1-\lambda)^2 s^2}{\pi} + \sigma_D^2\right].
\end{aligned}
$$

Primitives

- s: language-specific skill
- a: general specification ability
- ω: opportunity value
- b: activation cost
- ρ: risk aversion
- λ: delegated execution share

$$
V^g = \max_{m \in \mathcal{M}_g} V^m, \qquad Z^g = 1\{V^g \geq 0\}.
$$

Source: Quispe and Xu, Section 4.1, Equations (1)-(3).

Only delegation moves unfamiliar-language entry

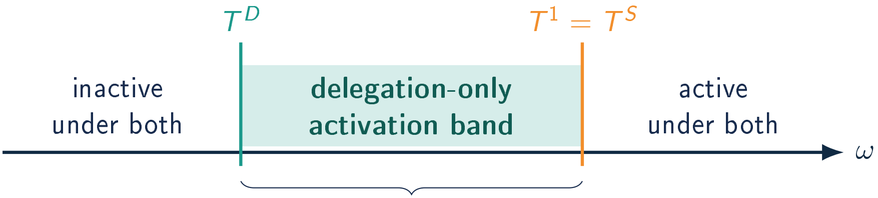
$$T^S = b - s\mu + \frac{\rho s^2}{2\pi},$$
$$T^1 = T^S - \max\{0, \gamma s - r_C\},$$
$$T^D = b - (1-\lambda)s\mu - \lambda a z(A) + \kappa + r_D + \frac{\rho}{2}\left[\frac{(1-\lambda)^2 s^2}{\pi} + \sigma_D^2\right].$$

Assumption 1 supplies the separation

For unfamiliar languages, $\gamma s - r_C \leq 0$, hence $T^1 = T^S$. Conversational augmentation does not move entry; delegation can if $B \equiv T^S - T^D > 0$.

Source: Quispe and Xu, Section 4.2, Equations (4)-(7), pp. 14-15.

Delegation activates a middle band of opportunities



$$Z^2 - Z^1 = 1\{T^D \leq \omega < T^S\}$$

Proposition 1: menu inclusion gives $Z^2 \geq Z^1$ path by path.

Proposition 2: with $B > 0$ and continuous F, activation probability is $F(T^S) - F(T^D) \geq 0$.

Source: Quispe and Xu, Section 4.2, Propositions 1-2, pp. 15-16.

The stock can keep growing after the first-use flow peaks

Proposition 3:

$$\Delta C_i(s) = \sum_{k \in \mathcal{U}_i} \left[(1 - p_{ik}^1)^{s+1} - (1 - p_{ik}^2)^{s+1} \right] \geq 0.$$

Flow	Stock
Newly used languages can spike at adoption and then fall as the at-risk set is depleted.	Cumulative languages retain first uses, so the gap can continue to rise.

Printed strict claim: under $p^1 = 0 < p^2$, the paper states strict increase and concavity over the observed horizon. The allowed endpoint $p^2 = 1$ breaks both strict inequalities.

Source: Quispe and Xu, Section 4.3, Proposition 3, p. 17; Theory Appendix proof.

At adoption, language breadth rises sharply

Outcome	$ATT(0)$	SE
Active languages	**+2.528**	0.063
Newly used languages	**+1.193**	0.051
Language entropy	**+0.382**	0.009
Cumulative languages	**+1.604**	0.054

5,346 developers; 28 months; 3.15 million commits; 57.2 million changed files.

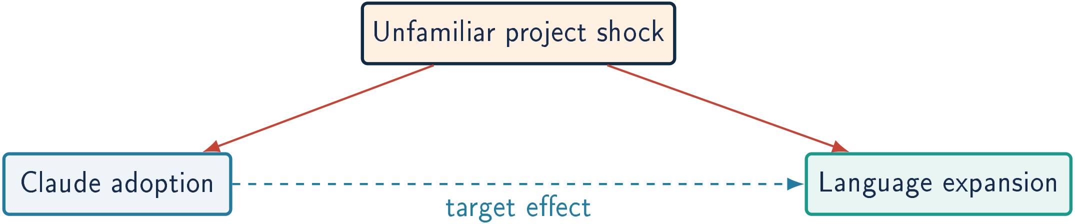
Economic magnitude

Pre-adoption active-language mean: **0.90**. The active set roughly triples on impact.

The cumulative series has non-trivial pre-trends and is treated as descriptive.

Source: Quispe and Xu, Sections 5 and 7; Table 2; source table `table_cl_main.tex`.

The event study does not identify causality



- Staggered DiD handles cohort heterogeneity and avoids already-treated comparisons.
- Removing Claude commits and the first-Claude language addresses mechanical measurement.
- Neither resolves project-driven adoption; activity already rises at $e = -1$.

Conclusion: robust event-time associations consistent with delegation, not a causal estimate.

Source: Quispe and Xu, Section 9, "Discussion: Identification Threats."

AI primitives imply expansion or deskilling

Quispe and Xu

AI adds specification and delegated execution:

$$\{S, C\} \longrightarrow \{S, C, D\}.$$

A lower threshold activates unfamiliar-language opportunities: the extensive margin expands.

Aouad, Lykouris, and Zhong

Skill, effort, and AI assistance are substitutable inputs:

$$x = s + e + a, \qquad \max_{e \geq 0} p(x) - \gamma e.$$

More AI can reduce effort; with endogenous skill or unreliable assistance, productivity can fall and skills can polarize.

Modeling contrast: menu expansion creates a feasible margin; input substitution can cannibalize effort and skill formation.

Source: Quispe and Xu, Sections 1 and 4; Aouad, Lykouris, and Zhong (2026), Sections 1-5, arXiv:2605.11350v1.

Lean requires $p_2 < 1$ for strict dynamics

Printed: $p_1 = 0 < p_2 \le 1 \Rightarrow G_s < G_{s+1}$ and $\Delta^2 G_s < 0$. **But at $p_2 = 1$:** $G_s = 1 - (1-1)^{s+1} = 1$.

Corrected Spec boundary

```
dynamicCumulativeLanguageEffectCorrectedSpec
  0 <= p1 <= p2 <= 1 -> weakGap
  p1 = 0 and 0 < p2 < 1
    -> strictIncrease and strictConcavity
```

Formal refutation

```
theorem ...EndpointCounterexample :
  not ...PrintedStrictClaim := by
  intro h
  have hUnit := h Unit (fun _ => 1) ...
  have hFalse := hIncreasing 0
  norm_num at hFalse
```

Five endpoints compile with zero ‘sorry‘, ‘admit‘, or axioms; build and ‘check –fast‘ succeed.
Verdict: partially formalized.

Source: `lean/PaperInterface.lean`; `MainTheorems.lean`; `ProofInterface.lean`; validation report.

The handwritten check makes saturation visible

Pending handwritten verification

`hand/prop3-endpoint.jpg`

Endpoint $p_2 = 1$ of Proposition 3

Two-minute derivation

Set one language, $p_1 = 0$, $p_2 = 1$. Then show:

$$G_s = (1-0)^{s+1} - (1-1)^{s+1} = 1.$$

Hence $G_{s+1} - G_s = 0$, contradicting strict growth; the second difference is also zero.

Source: Quispe and Xu, Proposition 3 and its Theory Appendix proof; independent Lean endpoint audit.

Expansion does not uniquely identify delegation

Extension: generic productivity

Replace skill-complementary augmentation with

$$V^P = V^S + q - r_P, \qquad T^P = T^S - (q - r_P).$$

If $q > r_P$, even a non-agentic shock creates an activation band. Expansion alone does not reveal whether T^P or T^D moved.

Discriminating test

Cross post-adoption change with:

- pre-adoption portfolio breadth,
- independent verification ability,
- task-level delegation intensity.

Delegation predicts a positive interaction; a common productivity shift does not.

Bottom line: a large shift in production breadth; five compiled Lean cores; one strict-endpoint defect; causal and mechanism-specific identification remain open.

Source: Quispe and Xu, Theory Appendix and Sections 8-10; extension developed in `extensions.md`.